

The Future Worth Building Is Human

Thinking Machines

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The mission of Thinking Machines is to build AI that extends human will and judgment.

Artificial intelligence can do more every day, but deciding what it should do is up to us: individuals, organizations, humanity as a whole. These decisions require knowledge and judgment that people acquire through continuous contact with the work, increasingly done alongside AI. Shaping the goals of advanced intelligence is also a continuous process of feedback, learning, and realignment.

Most AI in use today is trained in a handful of places and then frozen. It isn't shaped by the people it serves, and doesn't learn much from the work they do together.

To progress on that path, we are pursuing these technical directions:

- We train strong models, advancing capabilities such as multimodal interaction and customizability. Sharp instruments extend human will, and human judgment needs to shape models that compete on the frontier.
- We build tools that enable people to make AI their own, customizing models to serve their unique needs. This includes the ability to train model weights.
- We develop interfaces that broaden the communication channel between human and machine, allowing personal judgment to continuously influence the work of AI.
- We publish research for the scientific community, because the power to shape AI requires deep understanding of how it's made.

We believe the future worth building is human — shaped by human knowledge, guided by human will, and decided by human judgment. What follows is the case for that future, and the work we're doing to bring it about.

Bringing intelligence to knowledge

AI exists to serve the work that we do. This work runs on knowledge of how things are done and what is worth doing, knowledge that is generated continuously by people engaged in the work.

Think of a chef crafting a new recipe or a shopkeeper rearranging the items and prices on display. They are pursuing a complex set of goals and applying know-how that isn't immediately legible to outsiders. This knowledge is constantly updated through feedback; it's not a static repository that can be written into a database. It's local — a different restaurant or shop pursues different outcomes by different means. The collective knowledge of shops and kitchens is scattered across every shopkeeper and chef.¹

The dispersion of knowledge is a collective strength; it's the source of variety, adaptability, and resilience of the overall system. It's the reason that free markets

and held privately by those who acquired it through their work.² Attempting to aggregate knowledge for the use of a centralized intelligence faces the same challenge.

There are domains where intelligence alone is sufficient, and where autonomous AI doesn't require human participation to race ahead. Two examples are chess, where the strongest engines are trained purely on self-play, and math, where frontier models are solving long-standing problems on their own. These examples share two traits. First, the goal given to AI is static and expressible: to win a chess match, to prove a theorem. Second, these domains don't contain hidden knowledge. The rules of chess and math are universal; the board is visible to all. Outside the board, intelligence alone is not enough.

For artificial intelligence to benefit from distributed knowledge, it must itself be distributed. Every organization is powered by the expert knowledge of its people, gained and expressed through their work. We believe in AI that helps the organization cultivate that unique knowledge, not AI that extracts a snapshot of it and replaces it with a standard offering. This cultivation is an ongoing process that requires AI to work with people, not in their stead.

In 2014, Toyota, long a master of the automated plant, brought its expert craftsmen back onto the line with the explicit goal of growing craftsmanship and knowledge. The man who led this, Mitsuru Kawai, put the reason this way: "To be the master of the machine, you have to have the knowledge and the skills to teach the machine."³ The production of knowledge and application of intelligence lift each other; they are not substitutes.

The work people do may change, and turn toward more of what only people bring, but the best organizations will make the fullest use of both. AI should enable each organization to be excellent in its own way, not to erase the differences between them.

We aim to bring intelligence to where knowledge is made and used. We build tools that enable everyone to fine-tune models with their unique knowledge, and to keep adapting the models as their knowledge evolves. We publish research and recipes that put this capability within reach of more people. We envision frontier AI as a collective,

Human participation is a technical challenge

Keeping people engaged in setting goals and sharing knowledge with AI doesn't mean resisting automation for its own sake. What a machine does reliably on its own, it should do. But it should also know when to act alone and when to invite oversight and feedback, as people themselves do when working in teams. The best collaborators anticipate: they learn what someone is reaching for and bring it before being asked, earning over time the right to act on their behalf. These are technical challenges, requiring a new approach to how AI is designed and evaluated.

A major bottleneck for bringing human knowledge and judgment to work with LLMs is the communication channel between human and AI — a small text box and a long wait. This is too narrow to carry the richness of human wisdom and intent, and too slow for ongoing feedback. People collaborate best when they collaborate live. We interrupt and correct, take second looks and make gestures, change our minds aloud. This is why we're making a long-term bet on [interaction models](#): models that handle live, multimodal interaction natively, in the model itself rather than in scaffolding bolted around it. Built this way, interactivity scales with intelligence; the same training that makes the model smarter makes it a better collaborator. The right interface doesn't just allow human participation, it invites and rewards it.

Another challenge is setting the right target for evaluation and optimization. The common measure of AI intelligence today is the time horizon of software tasks models can execute autonomously, tracked on charts like METR's.⁴ We expect progress on this benchmark to continue, but it ultimately measures only what AI is capable of on its own, not what people and machines can accomplish together.

Measuring the latter is more complex, and can't be done by a lab on its own. Every organization evaluates for itself whether AI helps it sharpen its judgment, develop new knowledge, and achieve its objectives.

each user distinct and devaluing the cultivation of specialized knowledge. By optimizing AI to be customized and collaborated with, we benefit when our customers leverage their unique advantages. These advantages are maximized not by renting an AI and outsourcing to it, but by organizations owning it and tailoring it to their goals.

Decentralized alignment

Human values, just like human knowledge, reside in the heads of individual people and resist consolidation. But today, the values and voice of AI are decided in a handful of places. A single locus of value alignment, however well run, becomes a locus of power to be captured.

This creates danger, especially if most valuable work is done by AI on its own with little need for human input. The social contract between corporations, governments, and citizens relies on individuals' productive capabilities on which the government's sovereignty and corporations' profits ultimately depend. Power that needs nothing from people loses the incentive to care for their needs and values, caring instead for its own preservation.⁵

Even with the best intentions, a model shaped in one place inevitably encodes the values of its owner, not the individual users it serves.⁶ Today each lab trains its next flagship model by using its previous flagship model to generate training data and a reward signal. Whatever character emerges from that loop, everyone gets the same one, and each generation inherits the traits of the last, raised on its parent's outputs and judged by its parent's tastes. A single alignment spec suppresses creativity and diversity and stultifies progress. Free speech and free markets let new ideas, goods, and services emerge and compete, rather than averaging out the preferences that exist at a point in time.

For organizations and individuals to align AI to their own values, these values must be encoded in the model weights. If the user's values and desires only impact the model through a prompt, the user finds that surface properties change while the deeper

The power to shape a model profoundly is also the power to shape it for ill. John von Neumann remarked on this problem in 1955,⁸ writing that the useful and the harmful aspects of technology “lie everywhere so close together that it is never possible to separate the lions from the lambs.” Keeping the lambs safe is an ongoing process, the result of judgment exercised and choices made continuously. We aim to give the people making these choices stronger tools, pursuing research that enables safer models without taking away ownership.

Humanity has flourished through individual weirdness and creative tension. We envision alignment as a feature not of a single model but of an ecosystem of AIs raised in different places, disagreeing, competing, and learning from each other. We believe in keeping the weirdness alive.

The future worth building

The technology industry has made incredible progress in teaching machines to think; what they should think about must remain with us. What is worth wanting, what is worth making, what’s the right use of the time we have.⁹ We are not looking to hand down a single answer to this, but to give every person the ability to make their own answer part of the development of frontier AI.

The current path of AI development, pushing towards centralization and autonomy, frames human involvement as a trade-off: participation vs. capability, ownership vs. safe alignment. We see these as technical challenges to solve: AI that is more capable because it encourages human participation, organizations that benefit in the long run from tailoring AI to their advantages, alignment that arises from diverse AIs shaped by the people who own them. Solving these challenges is what our mission requires.

The future is not a choice between human dominance and rapid obsolescence in the face of AI. Different roads lead to many different futures, and we get to choose which one to take. We are building technology that lets the born and the made walk the road together.

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